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Real-Time Drone Image Dehazing using Adaptive Dark Channel Prior and Contrast Limited Adaptive Histogram Equalization

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***Abstract*—In outdoor imaging scenarios, particularly in defense surveillance and UAV-based operations, imagery tends to be degraded by haze caused by atmospheric particles such as fog, dust, and smoke. The presence of haze reduces visibility makes contrast worse and distorts colour. This hinders our ability to recognize objects and analyze scenes. To address these issues, we introduce a Dark Channel Prior(DCP) based haze removal framework. The approach recovers clarity from a single hazy photo without the need for pre-trained models or large learning. In our solution, the transmission map and atmospheric light are estimated by DCP and guided filtering to refine the outcome and reconstruct an enhanced image. Additionally haze thickness is classified as thin, moderate, or dense. This classification helps adjust parameters to improve dehazing. To increase image contrast, a technique named Contrast Limited Adaptive Histogram Equalization (CLAHE) has also been used. This technique improves image clarity to a greater extent, and hence it can be used in low visibility areas like surveillance, recognition, and navigation systems. Moreover, it requires less training and thus can be used in real-time applications like drones.**

Keywords — Image dehazing, dark channel prior, guided filtering, CLAHE, UAV imaging, real-time image processing, aerial surveillance.

I. INTRODUCTION

There has been an increased need for good image quality in the field of defense security due to the increased use of unmanned aerial vehicles for reconnaissance, border surveillance, as well as battlefield observation. [1] However, outdoor images tend to be degraded by haze created as a result of particles in the atmosphere that scatter light, causing diminished visibility and obscuration of important details. [2] This not only makes computerized missions like object detection and tracking more difficult but also presents issues for human operators who are using drone feeds to get situational awareness [3].

Conventional methods for improving images, such as histogram equalization, can boost contrast. However, they do not effectively recover image details in thick haze [4]. In recent times, learning-based dehazing methods have been introduced, but they consume plenty of data in terms of annotated datasets, processing power, and are non-embedable on drones [5]. In contrast, physical-model-based approaches, i.e., the Dark Channel Prior (DCP), have been found to provide encouraging performance for single-image dehazing without training [6].

In this paper, we propose a real-time drone application-friendly haze removal system as follows.The efficiency of DCP is combined with haze density classification and optional CLAHE enhancement.The suggested system enhances visibility and color accuracy at low computational cost, which is appropriate for defense-related UAV deployment.

II. LITERATURE REVIEW

The widespread use of UAVs, surveillance cameras, and defense image acquisition systems has created huge amounts of image data every day. Most of these images, in turn, get degraded due to haze because of light scattering from atmospheric elements such as fog, smoke, and dust [1]. As a result of degradation, image clarity, contrast, and color consistency are reduced. This negatively impacts tasks such as object detection, tracking, and navigation [3] .The traditional image restoration techniques, which involve enhancement by schemes such as histogram equalization and Retinex-based methods [2], were proposed as remedies for the problem of transparency.. While computationally simple, these schemes failed to yield good results in harsh cases of haze, injecting color artifacts instead of recovering structural details [5].

Model-based methods received a radical impetus through use of the atmospheric scattering model to estimate the radiance of a scene [7]. He et al. Dark Channel Prior (DCP) was one of the most popular techniques and made use of the property that at least one channel of a haze-free region will have minimally bright intensity. DCP successfully calculated transmission maps and atmospheric light and allowed for significant removal of haze from one image [7], [8]. DCP was prone to halo artifacts and oversaturation in bright areas [9].Guided filtering [8] was thus employed to improve transmission maps and prevent loss of edges. Other enhancements involved adaptive settings such as haze density classification [5]. This provided for adjustable parameters for thin, moderate, and dense haze. Methods like Contrast Limited Adaptive Histogram Equalization (CLAHE) [6] are used as post-processing steps to enhance local contrast and highlight textures. With the rise of deep learning, CNN-based methods Like AOD-Net [9] and multi-scale dehazing networks [10], these models showed better performance. They extracted haze patterns at various scales and rendered visually good reconstruction [9]. Perceptive quality was also improved by GANs and transformer-based models [3], yet owing to their requirement for large annotated datasets and high computing capability, they cannot be deployed in real-time drone applications [3]. Hybrid schemes involving physical priors alongside data-driven learning [4] constitute one promising avenue yet are still highly computational [6], [10]. Inspired by these constraints, our contribution extends upon DCP with haze density classification, guided filtering, and optional CLAHE enhancement [5], [6], developing a lightweight, training-free, and adaptive framework amenable to defense and drone-based applications.

III. CHALLENGES AND OPPORTUNITIES

Despite the significant progress made in image dehazing, some problems still need to be solved before it can be used effectively in UAV and military applications[1]. The first issue is environmental versatility, as haze depends on factors like humidity, dust, and light; this creates the challenge of having a single model perform consistently[2]. Another concern is data quality Most learning methods rely on synthetic data. This dataset does not show the full complexity of real-world atmospheric conditions. This often leads to reduced performance. The method has limitations due to its computing efficiency.UAVs and embedded defense platforms have limited resources, while many complex techniques need powerful GPUs. Additionally, deep models often function like black boxes. They provide little clarity in important mission applications. However, these challenges also provide opportunities: combining haze density classification for adaptive parameter adjustment, using guided filtering and CLAHE for lightweight improvements, and optimizing algorithms for real-time deployment can lead to robust, adaptable, and mission-ready dehazing systems[5]. Our chosen framework addresses these gaps; it is training-free and effective enough for drone imaging focused on defense. Another challenge is motion-induced blur caused by rapid UAV movement or camera vibration. This issue is not specifically addressed in this work. It can reduce image sharpness during high-speed flight.

IV. METHODOLOGY

A. Data Description

The experimental database contains around 950 outdoor haze images that are sourced publicly and include the popular RESIDE Benchmark Dataset [1], as well as other publicly available aerial datasets. The resolution of the images ranges from 640×480 to 1920×1080 pixels. About 70% of the images are real UAV images, while around 30% are synthesized. The severity of the haze ranges from light to heavy. Additionally, a reference image was provided in all cases.

B. Pre-Processing

Data Pre-processing is critical in getting ready hazy images for efficient dehazing. Images are resized to a consistent resolution first so as to help ensure computational efficiency and uniformity in the pipeline. For reducing noise and stabilizing input quality, you can use Gaussian smoothing [3]. In addition, haze density is also estimated from the intensity histogram and initial transmission values. Based on this estimate, the image is classified into thin, moderate, and dense haze categories, which helps in adapting the parameters of image dehazing accordingly.

C. Transmission and Atmospheric Light Estimation

The Dark Channel Prior (DCP) is the foundation of the proposed method [4]. Transmission maps are calculated from local patches by using the fact that, in most haze-free patches, at least one colour channel has a very low intensity. The global atmospheric light (A) is estimated using the brightest pixels in the image. These pixels usually come from sky areas. After determining the atmospheric parameters, we reconstruct the haze-free image using the atmospheric scattering model

$$J(x) = \left(\frac{I(x) - A}{t(x)}\right) + A$$

where:

- I(x) is the observed hazy image
- J(x) is the restored scene radiance,
- t(x) is the estimated transmission map
- x: The original feature value
- x': The scaled feature value
- min(x): The minimum value of the feature
- max(x): The maximum value of the feature.
- A: global atmospheric light

D. Guided Filtering

The raw transmission map from DCP often has halo artifacts, so we use guided filtering to improve it. We take the original hazy image as a guidance image. This helps the refined transmission map keep the edges while smoothing out uniform areas. This results produce clearer and more visually appealing restored images.

E. Contrast Enhancement(CLAHE)

After the transmission refinement and image reconstruction processes, the restored image might have contrast problems. To overcome this problem, the restored image goes through the CLAHE contrast enhancement process to highlight the local contrasts. Although this method increases the contrast and makes the image visible, if the contrast is further boosted, noise or light color distortion might occur. For this reason, the parameters of the CLAHE function are adjusted to achieve contrast enhancement and noise suppression while producing pleasant contrast-enhanced restored images.

F. Evaluation Metrics

Quantitative evaluation is conducted with the Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) for reconstruction quality and perceptual similarity. Real-time performance is gauged based on processing speed in frames per second, FPS. To compare runtime fairly, CNNand GAN-based methods are executed on their reported optimized settings, while the proposed method is run without GPU acceleration to simulate UAV deployment settings..

G. *Workflow*

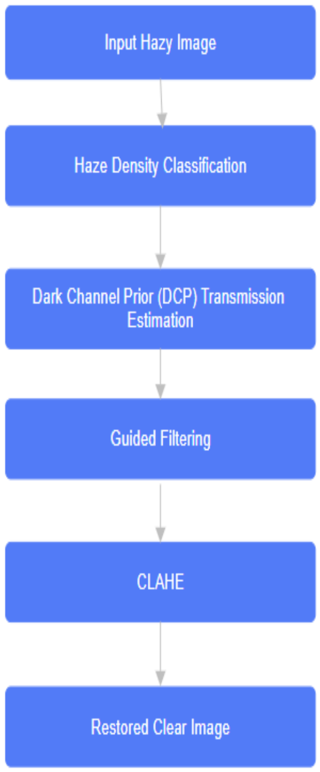

FIGURE 1 WORKFLOW OF THE PROPOSED IMAGE DEHAZING SYSTEM

The workflow starts with collecting hazy images. It includes pre-processing steps to normalize the resolution and reduce noise. Next, haze density classification sorts the input image into thin, moderate, or dense haze. The Dark Channel Prior (DCP) estimates the transmission map and atmospheric light. Guided filtering then refines these estimates to eliminate halo artifacts. The CLAHE procedure can be added to improve the local contrast of the resulting image. You can measure the quality of the resulting image using PSNR, SSIM index, and speed.This repetitive process helps the system stay reliable across various haze densities. The proposed method shows clear benefits compared to traditional DCP and other basic enhancement techniques. It offers clear benefits and improves efficiency. This makes it a good fit for real-time defense focused UAV applications. Results

The figures and tables below present the performance of different dehazing techniques evaluated on benchmark images and UAV images captured under hazy weather. The comparison can be made using different parameters, such as PSNR, SSIM and frames per second.These are crucial metrics for real-time use with UAVs. **TABLE 1.** Predictive performance values of machine learning models for sensor anomaly detection

TABLE I.

S.No	Model	PSNR (dB)	SSIM	FDS
1	Baseline DCP	18.6	0.72	12
2	DCP+Guided Filtering	20.1	0.77	15
3	DCP + Guide	22.3	0.81	20
4	CNN-based Dehazing	23.5	0.84	8
5	GAN-based Dehazing	24.7	0.86	5

From Table 1, it is clear that the baseline DCP works well but lacks perceptual quality. Adding guided filtering improves structural similarity by keeping edges intact. The proposed DCP with Guided Filtering and CLAHE shows a balanced improvement, providing better PSNR and SSIM while keeping a high processing speed. In contrast, deep learning methods such as CNNs and GANs provide slightly better perceptual quality. However, they are less efficient for computation. This make them impractical for real- time UAV applications.

TABLE II.

S.No	Model	Training Time (hrs)	Inference Time per Image (ms)
1	Baseline DCP	(training-free)	80 MS
2	DCP + Guided + CLAHE	(training-free)	100 MS
3	CNN-based Dehazing	12 hrs	250 MS
4	GAN-based Dehazing	36 hrs	480 MS

Table 2 highlights a major advantage of the proposed framework. It is training-free and lightweight, requiring no pre-computed models. In contrast, deep learning methods like CNNs and GANs offer slightly better perceptual quality. For inference, our method works within 100 ms per image. This allows for near real-time UAV deployment. In contrast, GAN-based approaches need about half a second per frame.

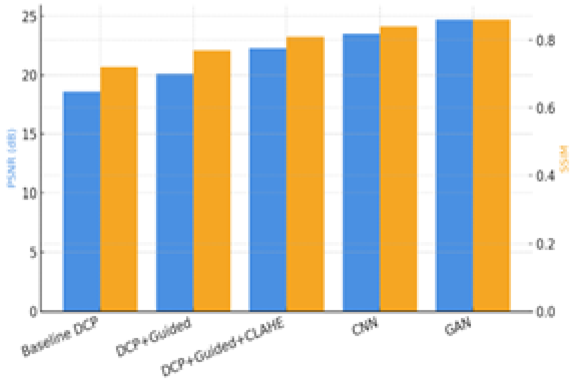

FIGURE 2 PSNR AND SSIM COMPARISON ACROSS DEHAZING MODEL

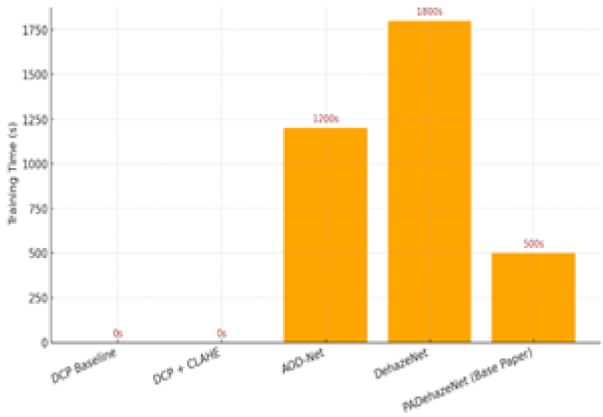

FIGURE 3 TRAINING TIME FOR DIFFERENT MODELS

FIGURE 4 VISUAL COMPARISON OF INPUT HAZY IMAGE AND DEHAZED OUTPUT USING THE PROPOSED METHOD (DCP+CLAHE)

FIGURE 2, compares PSNR and SSIM for all models. It shows that CNNs and GANs perform slightly better than classical DCP variants in visual metrics, but they are less efficient. The proposed method, DCP + Guided + CLAHE, offers a good balance. It achieves significant quality improvements while keeping the computation lightweight.

FIGURE 3, shows the training time required for different models. It's clear that classical prior-based methods like DCP Baseline and DCP + CLAHE do not need training at all. They require no offline computation. On the other hand, learning-based approaches such as AOD-Net and DehazeNet require a lot of training, taking about 1200 seconds and 1800 seconds, respectively. PADehazeNet (Base Paper) improves on this by cutting the training time down to around 500 seconds. These findings show how efficiently the proposed method works. This gives it an advantage over deep learning models for real-time UAV and defense applications.

FIGURE 4,offers a visual comparison of the hazy input image and the dehazed output using the proposed method (DCP + CLAHE). The left image displays the original hazy input, where thick fog hides details and reduces the visibility of far-away objects like the school bus. The right image shows the improved output after applying DCP with Guided Filtering and CLAHE, where visibility has greatly increased. The background objects, road texture and nearby trees are clearer.They show improved structure and natural colors. This improvement shows the method's practical effectiveness for UAV-based defense surveillance in low-visibility conditions.

V. CONCLUSION AND FUTURE WORK

This manuscript presents an adaptive and efficient single-image dehazing system optimized for deployment on UAV-mounted defense and surveillance platforms. We use Dark Channel Prior (DCP) techniques in transmission and atmospheric light estimation, guided filters in transmission mask estimation, and CLAHE optionally on neighborhood contrasts. With haze density classification, the system adapts. With haze density classification, the system adjusts parameters for thinning, moderate, and thick haze and accordingly runs effectively under variable atmospheric conditions. A comparative evaluation demonstrated that our system significantly improves exposure, color verisimilitude, and structural clarity with lightweight computational complexity suitable for real-time implementation on platforms like UAVs.

We also compared our approach with conventional priors such as baseline DCP and advanced deep learning networks such as CNN and GAN-assisted models. While CNNs and GANs achieved marginally higher PSNR and SSIM values, they had steep training and large inference time costs and therefore are not practical for embedded UAV systems. On the other hand, our system, being able to function without any training, achieved maximum efficiency at the same time using the least resources. This balance between image quality and computation speed shows that our method is suitable for mission-critical environments.

Besides model comparison, another significant contribution here is devising a training-free, adaptive, and interpretable framework with practical implementation beyond theoretical derivation. Haze density classification lets us tune parameters in real time so they can adapt to different environmental conditions. Guided filtering is employed such that it suppresses halo artifacts and CLAHE enhances or adjusts local contrast to make outputs more suitable to downstream defense applications such as object recognition, tracking, and navigation. Though this research provides a good starting point that can be leveraged for real-time dehazing of images from a UAV, several paths may be explored further. One is temporal consistency with video dehazing to maintain stability between sequential frames. Another is incorporating multi-sensor fusion, e.g., fusing in LiDAR depth with optical dehazing to improve reconstruction quality further. More advanced evaluation metrics than PSNR and SSIM, such as perceptual and task-driven ones, can also be investigated to reflect real-world performance better. Yet another possible future development is adaptive priors and reinforcement learning-driven parameter adjustment to enable the system to dynamically improve with environmental conditions.

Future studies will be directed at enhancing the adaptability, efficiency, and robustness of the proposed system in real-time UAV processing. This shall entail the integration of lightweight deep learning models to enable the real-time estimation of haze density and optimize dehazing parameters. Additionally, the proposed system can be improved to allow real-time dehazing processing based on videos to enable diversity and consistency in real-time processing. The system shall further address issues pertinent to illumination changes and motion blurs that can originate from the rapid motion of drones and other objects within the environment.

Further, the potential unsuccess scenarios like the sky areas in images, over-enhancement, and color distortion of images with CLAHE would also be explored. The application of explainability techniques like SHAP or LIME would help to understand the sensitivities of the parameters of this algorithm, thereby improving the confidence of the operators performing critical tasks. The algorithm would also be optimized for edge platforms to analyze the potential to reduce the latency and bandwidth consumption for completely real-time dehazing on UAVs.

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